Linear Algebra I Recitation Class: Week 6

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Throughout these notes, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ denote the sets of rational, real, and complex numbers, respectively. V is a vector space over a field F. If you are not yet familiar with fields, you may keep the examples $F = \mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ in mind.

Let V (resp. W) be a finite dimensional vector space with an ordered basis $\beta = \{v_1, \cdots, v_n\}$ (resp. $\gamma = \{w_1, \cdots, w_m\}$). We have the following dictionary:

Vector space side (abstract)		Matrix side (concrete)
$v \in V, w \in W$	$\xrightarrow{\text{fix ordered bases } \beta,\ \gamma}$	$[v]_\beta \in F^n$, $[w]_\gamma \in F^m$
$T : V \to W$ linear transformation	$\xrightarrow{\text{fix ordered bases } \beta,\ \gamma}$	$[T]^\gamma_\beta \in M_{m\times n}(F)$, $[T(v)]_\gamma = [T]^\gamma_\beta [v]_\beta$

Therefore, everything we have learned and will learn about linear transformations between finite-dimensional vector spaces can be translated into statements about matrices. We will explore more about the dictionary in the future.

Example 1. Let $\mathbb{R}[x]_n$ be the vector space of polynomials in $\mathbb{R}$ with degree not larger than n, then $\mathbb{R}[x]_n$ has an ordered basis $\beta_n = \{1, x, x^2 \cdots, x^n\}$. Define the map

$$T : \mathbb{R}[x]_3 \longrightarrow \mathbb{R}[x]_2 \qquad f(x) \longmapsto f'(x) + f''(x).$$

One can check that T is linear. Let us compute $[T]^{\beta_2}_{\beta_3}$: $\beta_3 = \{1, x, x^2, x^3\}$ and

$$\begin{aligned} T(1) &= 0 + 0 = 0 + 0x + 0x^2 \\ T(x) &= 1 + 0 = 1 + 0x + 0x^2 \\ T(x^2) &= 2x + 2 = 2 + 2x + 0x^2 \\ T(x^3) &= 3x^2 + 6x = 0 + 6x + 3x^2 \end{aligned}$$

Therefore,

$$[T]^{\beta_2}_{\beta_3} = \begin{pmatrix} 0 & 1 & 2 & 0 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

In the rest of the notes, we investigate subspaces that "behave well" under the action of linear transformation $T : V \to V$:

Definition 2. Let $T : V \to V$ be linear. A subspace W of V is called **T-invariant** if $T(W) \subseteq W$.

There are some immediate examples of invariant subspaces, but those are not quite of our interest:

Exercise A. Prove that $\{0_V\}, V, \operatorname{im} T, \ker T$ are all T-invariant.

Let W be a T-invariant subspace of V, then there are two specific maps induced from T. The first one is the **restriction of T to W**, it is defined to be the map $T_W : W \to W$ by $T_W(x) = T(x)$ for all $x \in W$. One can easily check that T_W is a linear transformation since it is essentially T, which is linear. The only difference is in their domains and codomains.

The second map is defined on the quotient space V/W by

$$\overline{T} : V/W \longrightarrow V/W \qquad v + W \longmapsto T(v) + W$$

The map is well-defined: If $v + W = v' + W$, then $v - v' \in W$, and

$$T(v) - T(v') = T(v - v') \in T(W) \subseteq W.$$

Therefore, $T(v) + W = T(v') + W$. $\overline{T}$ is also linear since

$$\begin{aligned}\overline{T}(a(v_1 + W) + (v_2 + W)) &= \overline{T}((av_1 + v_2) + W) \\ &= T(av_1 + v_2) + W \\ &= [aT(v_1) + T(v_2)] + W \\ &= a[T(v_1) + W] + [T(v_2) + W] = a\overline{T}(v_1) + \overline{T}(v_2)\end{aligned}$$

for any $a \in F$, $v_1, v_2 \in V$. One can prove the general result in the similar manner:

Exercise B. Let V_1, V_2 be vector spaces and $W_1 \subseteq V_1, W_2 \subseteq V_2$ be subspaces. If $T : V_1 \to V_2$ is linear and $T(W_1) \subseteq W_2$, then the map

$$\overline{T} : V_1/W_1 \longrightarrow V_2/W_2 \qquad v + W_1 \longmapsto T(v) + W_2$$

is a well-defined linear transformation.

We will show that when V is finite dimensional, the matrix representation of T_W and $\overline{T}$ is related to T if we choose suitable bases. That is, let $\gamma = \{w_1, \cdots, w_n\}$ be an ordered basis for W which extends to a basis $\beta = \{w_1, \cdots, w_n, v_1, \cdots, v_m\}$ for V, then

$$\eta := \{v_1 + W, v_2 + W, \cdots, v_m + W\}$$

is an ordered basis for V/W. Since W is T-invariant, $T(w_i) \in W$ for any $i = 1, \cdots, n$, i.e., we can write

$$T(w_i) = a_{1i}w_1 + a_{2i}w_2 + \cdots + a_{ni}w_n \quad a_{1i}, \cdots, a_{ni} \in F \text{ for each } i = 1, \cdots, n$$

For v_i, $i = 1, \cdots, m$, we naturally have

$$T(v_i) = b_{1i}w_1 + b_{2i}w_2 + \cdots + b_{ni}w_n + c_{1i}v_1 + \cdots + c_{mi}v_m \quad b_\bullet, c_\bullet \in F.$$

Therefore, the matrix representation of T with respect to the ordered basis β is

$$[T]_\beta = \begin{pmatrix} a_{11} & \cdots & a_{1n} & b_{11} & \cdots & b_{1m} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} & b_{n1} & \cdots & b_{nm} \\ 0 & \cdots & 0 & c_{11} & \cdots & c_{1m} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & c_{m1} & \cdots & c_{mm} \end{pmatrix} = \begin{pmatrix} A & B \\ O & C \end{pmatrix}$$

Furthermore, the matrix representation of T_W with respect to the ordered basis γ is

$$[T_W]_\gamma = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} = A$$

In other words, $[T_W]_\gamma$ is a submatrix of $[T]_\beta$ and

$$[T]_\beta = \begin{pmatrix} [T_W]_\gamma & B \\ O & C \end{pmatrix}.$$

On the other hand,

$$\begin{aligned} \overline{T}(v_i + W) &= T(v_i) + W \\ &= (b_{1i}w_1 + b_{2i}w_2 + \cdots + b_{ni}w_n + c_{1i}v_1 + \cdots + c_{mi}v_m) + W \\ &= (c_{1i}v_1 + \cdots + c_{mi}v_m) + W \\ &= c_{1i}(v_1 + W) + \cdots + c_{mi}(v_m + W). \end{aligned}$$

Hence $[\overline{T}]_\eta = \begin{pmatrix} c_{11} & \cdots & c_{1m} \\ \vdots & \ddots & \vdots \\ c_{m1} & \cdots & c_{mm} \end{pmatrix}$. $[\overline{T}]_\eta$ is also a submatrix of $[T]_\beta$ and

$$[T]_\beta = \begin{pmatrix} A & B \\ O & [\overline{T}]_\eta \end{pmatrix}$$

To summarize, we have the following correspondence:

Vector space side (abstract)		Matrix side (concrete)
$T_W : W \to W$ $\overline{T} : V/W \to V/W$	$\xrightarrow{\text{fix ordered bases } \beta,\ \gamma,\ \eta}$	$[T_W]_\gamma, [\overline{T}]_\eta$ is a submatrix of $[T]_\beta$ and $[T]_\beta = \begin{pmatrix} [T_W]_\gamma & B \\ O & [\overline{T}]_\eta \end{pmatrix}$

In particular, if $V = W \oplus U$ and W, U are both T-invariant, let γ be an ordered basis for W, δ be an ordered basis for U, then $\beta = \gamma \coprod \delta$ is an ordered basis for V and the matrix representation of T with respect to β is

$$[T]_\beta = \begin{pmatrix} [T_W]_\gamma & O \\ O & [T_U]_\delta \end{pmatrix}.$$

The result can be generalized as the following:

Theorem 3. *Let V be a finite dimensional vector space, $T : V \to V$ be linear. If $W_1, \cdots, W_n$ are T-invariant subspaces and $V = W_1 \oplus W_2 \oplus \cdots \oplus W_n$, let β_i be an ordered basis for W_i, then*

$$\mathcal{B} = \beta_1 \coprod \beta_2 \coprod \cdots \coprod \beta_n$$

is an ordered basis for V and

$$[T]_{\mathcal{B}} = \begin{pmatrix} [T_{W_1}]_{\beta_1} & & & & \\ & [T_{W_2}]_{\beta_2} & & & \\ & & \ddots & & \\ & & & [T_{W_{n-1}}]_{\beta_{n-1}} & \\ & & & & [T_{W_n}]_{\beta_n} \end{pmatrix}$$

This matrix representation of T is of utmost importance since it simplifies an arbitrary matrix to a matrix with diagonal block submatrices, which is easier to understand and compute. In practice, we study the restriction of T to each invariant subspace W_i separately and then combine the resulting information to understand T on the whole space V. An important example of T-invariant subspaces is the **eigenspace E_λ corresponding to λ**, defined by

$$E_\lambda := \{v \in V \mid T(v) = \lambda v\}.$$

Eigenspaces play a vital role in the diagonalization of square matrices, which we will learn more about in the next semester.

Exercise C. Let V be a vector space and $T : V \to V$ be linear.

1. Prove that E_λ is a T-invariant vector subspace of V for any $\lambda \in F$.
2. Prove that $E_\lambda \cap E_\eta = \{0\}$ for any $\lambda \neq \eta \in F$.

Exercise D. (Challenging!) Let V be a vector space and $T : V \to V$ be linear. For any $\lambda \in F$, define the **generalized eigenspace corresponding to λ** by

$$K_\lambda := \{v \in V \mid (T - \lambda \operatorname{id}_V)^n(v) = 0 \text{ for some } n \in \mathbb{N}\}.$$

Note that $E_\lambda \subseteq K_\lambda$ since any $v \in E_\lambda$ satisfies $(T - \lambda \operatorname{id})^1(v) = T(v) - \lambda v = 0$.

1. Prove that K_λ is a T-invariant subspace of V for any $\lambda \in F$.
2. Prove that $K_\lambda \cap K_\eta = \{0\}$ for any $\lambda \neq \eta \in F$. (Hint: Prove first that K_λ is also $(T - \eta \operatorname{id}_V)$ invariant and its restriction $(T - \eta \operatorname{id}_V)_{K_\lambda}$ is 1-1.)

Remark. Let $T : V \to V$ be linear, we can define an $F[x]$ action on V by

$$\begin{aligned} F[x] \times V &\longrightarrow V \\ (f(x), v) &\longmapsto f(T)(v) \end{aligned}$$

where for

$$f(x) = a_n x^n + \cdots + a_1 x + a_0,$$

we define

$$f(T) := a_n T^n + \cdots + a_1 T + a_0 \operatorname{id}_V .$$

The $F[x]$-action combining the vector addition on V gives V an $F[x]$-module structure.

From this perspective, a subspace W is T-invariant precisely when it is closed under the action of all polynomials in T. Thus T-invariant subspaces are exactly the $F[x]$-submodules of V. Although we will not pursue this viewpoint further, it leads to a deeper theory that explains many results in linear algebra such as Jordan canonical form.

References

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